

1 Introduction

Each musical instrument's quality of sound—its timbre—is differentiated by the varying composition of harmonic frequencies ($f_n = nf_1$ for $n = 1, 2, 3, \dots$), which are present in addition to the fundamental frequency f_1 . In addition to the fundamental variation between instruments, the timbre also varies based on how notes are played (Perov, Polievkt et al., 2016).

Likewise, this has been my experience as a guitarist. Tone control is a major technical skill. Different compositions of harmonic overtones separate each musical instrument's sound from each other, and these harmonic overtones depend on where the string is plucked; plucking near the bridge will create a harsher, “tinny” sound while plucking toward the bridge will create a fuller sound.

1.1 Research Question

This investigation will explore how the quality of sound may change depending on where the string is plucked. The research question I have decided to investigate is: *How does changing the position of the plectrum along a guitar scale affect the frequency of the harmonic with the highest intensity when a string with constant tension is plucked?*

1.2 Background

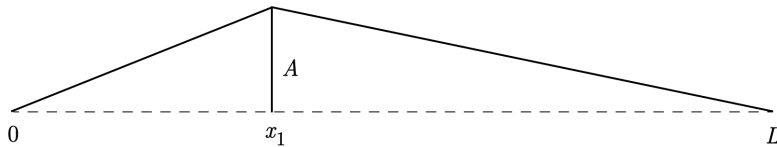


Figure 1: A guitar string with two fixed ends

A guitar string acts as a string with two fixed ends, where standing waves are generated as a string is plucked. Harmonics exist on the guitar string, however, the relationship between the intensity of each harmonic frequency is related to the quality of the sound. This investigation will look at how the frequency of the strongest harmonic (not the fundamental) changes as the position of plucking changes.

$$\sum_{n=0}^{100} \left(\lim_{x \rightarrow n} x^2 + 2x + 1 \right) \quad (1)$$

1.3 Hypothesis

According to Equation 1 in the background, it is hypothesized that

2 Methodology

2.1 Variables

	Area	Parameters
test	$\pi h \frac{D^2 - d^2}{4}$ (2)	h : height D : outer radius d : inner radius
test	$\frac{\sqrt{2}}{12} a^3$ (3)	a : edge length

Table 1: Variables

2.1.1 Pilot Study

2.2 Apparatus & Experimental Setup

- 6 stringed electric guitar (customized 6 string *Kiesel Guitars “Osiris”* model used)
- 1 set of 0.009 – 0.042mm gauge nylon electric guitar strings (*D’Addario XS Nickel “XSE0942 Super Light Coated Electric Guitar Strings”* model used)
- 1 audio interface with instrument input (*Audient EVO4* used)
- 1 instrument cable
- Computer with audio analysis software (*MATLAB* used)
- Tape measure
- Clamp stand

Figure 2 presents the experimental setup used to collect data. The guitar is clamped down for ease of plucking, and the tape measure is placed directly on the 2nd and 3rd strings to be on the same axis as the 1st string (which is plucked).



Figure 2: Experimental setup

2.3 Experimental Procedure

2.4 Safety, Ethical, and Environmental Considerations

3 Analysis

3.1 Raw Data and Observations

3.1.1 Raw Data

3.1.2 Observations

3.1.3 Fourier Analysis

3.2 Processed Data

3.3 Calculations and Uncertainty

3.4 Graphical Analysis

4 Conclusion

4.1 Scientific Reasoning

5 Evaluation

5.1 Strengths

5.2 Limitations

5.3 Further Study

References

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